

Homework 1: Matrix Algebra

Manual Solutions and NumPy Verification

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Code repository: github.com/Orlandoxinchen/homework1-linear-algebra

Part A. Conceptual / Manual Problems

1. Basic Operations

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}.$$

(a) **Addition and subtraction.** Elementwise computation gives

$$A + B = \begin{bmatrix} 1+2 & 2+0 \\ 3+1 & 4+2 \end{bmatrix} = \boxed{\begin{bmatrix} 3 & 2 \\ 4 & 6 \end{bmatrix}},$$

$$A - B = \begin{bmatrix} 1-2 & 2-0 \\ 3-1 & 4-2 \end{bmatrix} = \boxed{\begin{bmatrix} -1 & 2 \\ 2 & 2 \end{bmatrix}}.$$

(b) **Scalar multiplication.**

$$2A = \boxed{\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}}.$$

(c) **Matrix products.**

$$AB = \begin{bmatrix} 1(2) + 2(1) & 1(0) + 2(2) \\ 3(2) + 4(1) & 3(0) + 4(2) \end{bmatrix} = \boxed{\begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}},$$

$$BA = \begin{bmatrix} 2(1) + 0(3) & 2(2) + 0(4) \\ 1(1) + 2(3) & 1(2) + 2(4) \end{bmatrix} = \boxed{\begin{bmatrix} 2 & 4 \\ 7 & 10 \end{bmatrix}}.$$

Thus, $AB \neq BA$. Matrix multiplication is generally not commutative; changing the order changes which rows are combined with which columns.

2. Properties of Special Matrices

The following are four different examples:

$$\begin{aligned} \text{Diagonal: } \begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{bmatrix}, & \quad \text{Identity: } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \\ \text{Symmetric: } \begin{bmatrix} 2 & -1 \\ -1 & 4 \end{bmatrix}, & \quad \text{Idempotent: } \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

The symmetric example equals its transpose. The idempotent example satisfies $P^2 = P$.

3. System of Equations

The system

$$2x + y = 5, \quad 3x - y = 4$$

has matrix form

$$\underbrace{\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 5 \\ 4 \end{bmatrix}}_{\mathbf{b}}.$$

Since $\det(A) = 2(-1) - 1(3) = -5 \neq 0$,

$$A^{-1} = \frac{1}{-5} \begin{bmatrix} -1 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1}\mathbf{b} = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{bmatrix} \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{9}{5} \\ \frac{7}{5} \end{bmatrix}.$$

Hence $(x, y) = (1.8, 1.4)$.

4. Determinant and Rank

For

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix},$$

cofactor expansion along the first row gives

$$\begin{aligned} \det(C) &= 1 \begin{vmatrix} 1 & 1 \\ 5 & 7 \end{vmatrix} - 2 \begin{vmatrix} 1 & 1 \\ 3 & 7 \end{vmatrix} + 3 \begin{vmatrix} 1 & 1 \\ 3 & 5 \end{vmatrix} \\ &= (7 - 5) - 2(7 - 3) + 3(5 - 3) = 2 - 8 + 6 = \boxed{0}. \end{aligned}$$

Also, $R_3 = 2R_1 + R_2$, so the three rows are linearly dependent. The first two rows are not scalar multiples, so they are independent. Therefore,

$$\boxed{\text{rank}(C) = 2}.$$

The rank tells us that only two rows (and two columns) provide independent information. The matrix maps vectors into a two-dimensional subspace and is singular, consistent with its zero determinant.

5. Matrix Inverse

Let

$$D = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Its determinant is -1 , so it is invertible. Row reduction of $[D \mid I]$ (or the adjugate formula) gives

$$D^{-1} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & -2 & 2 \\ -1 & 2 & -1 \end{bmatrix}.$$

Indeed,

$$DD^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Part B. Python Verification

The required checks were performed with NumPy. The scripts are separated by question so each component can be run and reviewed independently.

Question 1: q1_basic_operations.py

```

1 import numpy as np
2
3 A = np.array([[1, 2], [3, 4]], dtype=int)
4 B = np.array([[2, 0], [1, 2]], dtype=int)
5
6 print("A + B =\n", A + B)
7 print("\nA - B =\n", A - B)
8 print("\n2A =\n", 2 * A)
9 print("\nAB =\n", A @ B)
10 print("\nBA =\n", B @ A)
11 print("\nAB equals BA:", np.array_equal(A @ B, B @ A))

```

Question 3: q3_system.py

```

1 import numpy as np
2
3 A = np.array([[2.0, 1.0], [3.0, -1.0]])
4 b = np.array([5.0, 4.0])
5
6 A_inverse = np.linalg.inv(A)
7 solution = A_inverse @ b
8
9 print("A inverse =\n", A_inverse)
10 print("\n[x, y] =", solution)
11 print("\nCheck A @ solution =", A @ solution)

```

Question 4: q4_determinant_rank.py

```

1 import numpy as np
2
3 C = np.array([[1.0, 2.0, 3.0],
4               [1.0, 1.0, 1.0],
5               [3.0, 5.0, 7.0]])
6
7 determinant = np.linalg.det(C)
8 rank = np.linalg.matrix_rank(C)
9
10 print("det(C) =", round(determinant))
11 print("rank(C) =", rank)
12 print("Row 3 equals 2*Row 1 + Row 2:",
13       np.allclose(C[2], 2 * C[0] + C[1]))

```

Question 5: q5_inverse.py

```

1 import numpy as np
2
3 D = np.array([[2.0, 1.0, 0.0],
4               [1.0, 1.0, 1.0],
5               [0.0, 1.0, 1.0]])
6
7 D_inverse = np.linalg.inv(D)
8
9 print("D inverse =\n", D_inverse)
10 print("\nCheck D @ D_inverse =\n",
11       np.round(D @ D_inverse, 10))

```

Verification Summary

Question	NumPy operation	Verified result
1	Array arithmetic and @	$AB \neq BA$
3	<code>np.linalg.inv</code>	$(x, y) = (1.8, 1.4)$
4	<code>det</code> , <code>matrix_rank</code>	$\det(C) = 0$, $\text{rank}(C) = 2$
5	<code>np.linalg.inv</code>	$DD^{-1} = I_3$

The complete source code and this report are available at <https://github.com/Orlandoxinchen/homework1-linear-algebra>.